BLINKIT SALES, DELIVERY AND CUSTOMER ANALYTICS

Sector: Quick Commerce
Institute: Newton School of Technology
Course: BTech Computer Science (AI/ML), 2nd Year

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Project Title: Blinkit Data Visualization and Analytics
GitHub Repository: click here!

Tableau Dashboard: click here!

Submission Date: 29-04-2026

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1. Executive Summary

This project studies Blinkit's business and operations using data. The goal is to understand how revenue, delivery performance, and customer satisfaction are connected.

We used a dataset from Kaggle and cleaned it using Python. After preparing the data, we analysed it and built dashboards in Tableau to show the results clearly.

The main finding is that Blinkit earns good revenue, but delivery performance is weak. About 62% of orders are delayed, and customer ratings are only average (around 3.3). Also, a large part of customer feedback is negative.

To solve this, we suggest improving delivery during peak hours, fixing slow stores and delivery partners, and improving customer experience. These changes can improve satisfaction and increase long-term growth.

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2. Sector and Business Context

Blinkit works in the quick commerce sector, where orders are delivered very fast, usually within minutes.

In this sector:

- Speed is very important
- Customers expect quick and smooth service
- Delays can lead to bad experience

This project helps business teams like operations managers and analysts understand where problems exist and how to fix them.

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3. Problem Statement and Objectives

The main problems are that Blinkit has the following:

- High delivery delays
- Average customer satisfaction
- Operational inefficiencies

Objectives:

- Study revenue and order trends
- Understand delivery delays
- Identify slow stores and delivery partners
- Analyze customer ratings and feedback
- Suggest improvements

Success Criteria:

- Clear identification of problems
- Strong connection between delay and satisfaction
- Useful business recommendations

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4. Data Description

Source: Kaggle Blinkit Dataset

Link: Click here!

The dataset includes:

- Orders and order details
- Customer information
- Delivery data
- Feedback and ratings
- Product and category data

Size:

- Around 50,000 records
- 30+ columns after processing

Limitations:

- Data is simulated
- No real-time updates
- External factors like traffic are not included

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5. Data Cleaning and ETL Process

We used Python to clean and prepare the data.

Steps:

- Removed duplicate rows
- Filled or handled missing values
- Converted date and time columns
- Merged multiple datasets

New columns created:

- Delay Minutes
- Delay Category (0–5, 5–15, 15+)
- Delivery time
- Peak vs Off-peak

Assumptions:

- Delay = actual time − promised time
- Peak hours based on high order volume

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6. KPI and Metrics

Business Metrics:

- Total Revenue
- Total Orders
- Average Order Value

Operational Metrics:

- % Delayed Orders
- Average Delay Time
- Delivery Time

Customer Metrics:

- Average Rating
- Repeat Customers
- Feedback

These metrics help understand performance from different angles.

7. Exploratory Data Analysis (EDA)

Revenue:

Revenue changes over time, showing some seasonal trends.

Delivery:

Delays increase during peak hours.

Customers:

Customers are evenly spread across segments.

Ratings:

Most ratings are between 3 and 4.

Insight:

Delivery problems are worse when demand is high.

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8. Statistical Analysis

We used basic statistical methods to confirm our findings.

- Correlation shows delay affects ratings
- Regression shows delivery time impacts performance
- Hypothesis tests confirm delayed orders behave differently

Conclusion:

Delivery delay is a key reason for lower customer satisfaction.

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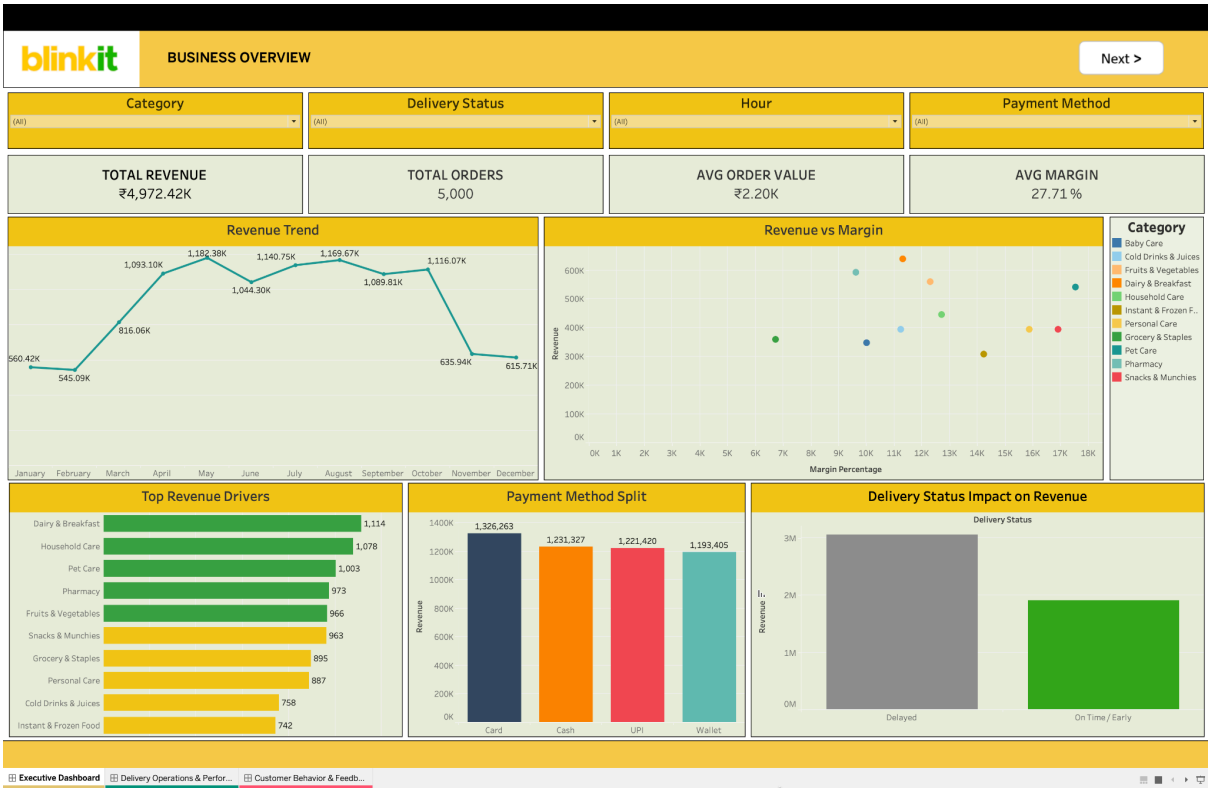
9. Dashboard Design

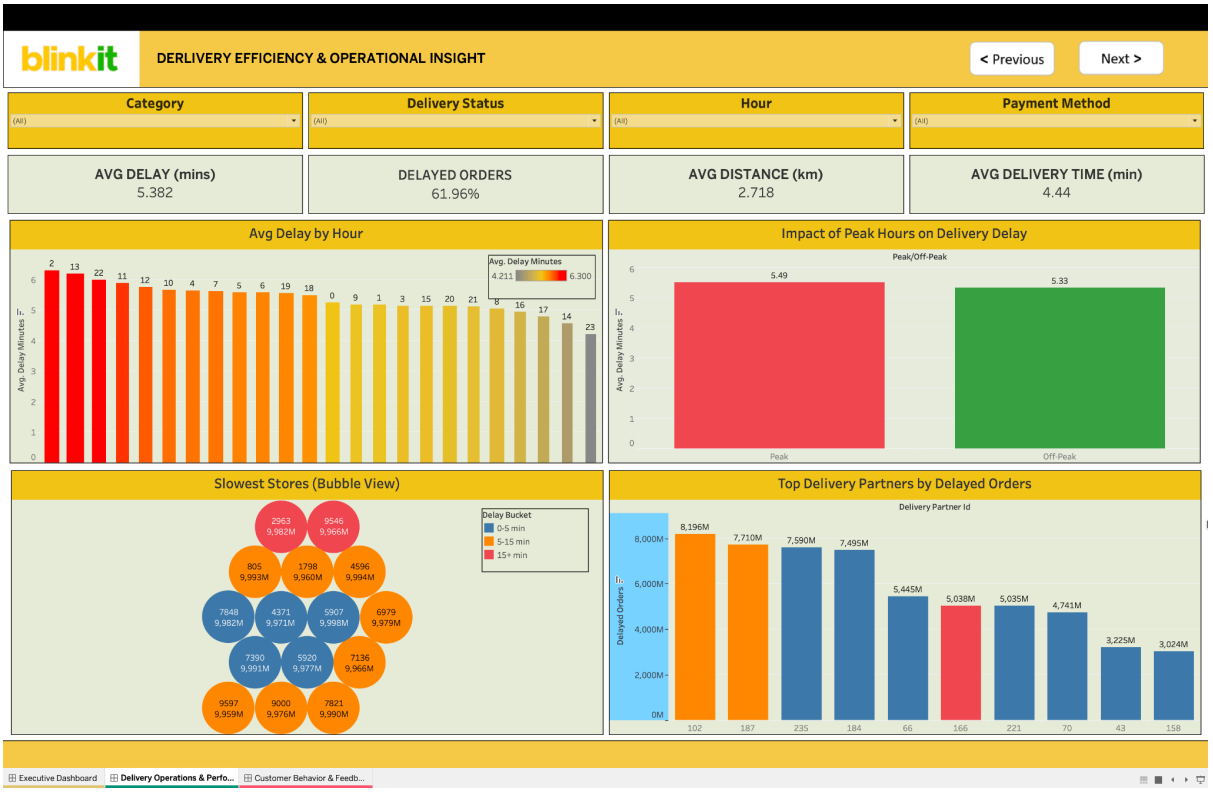
We created three dashboards:

Business Dashboard:



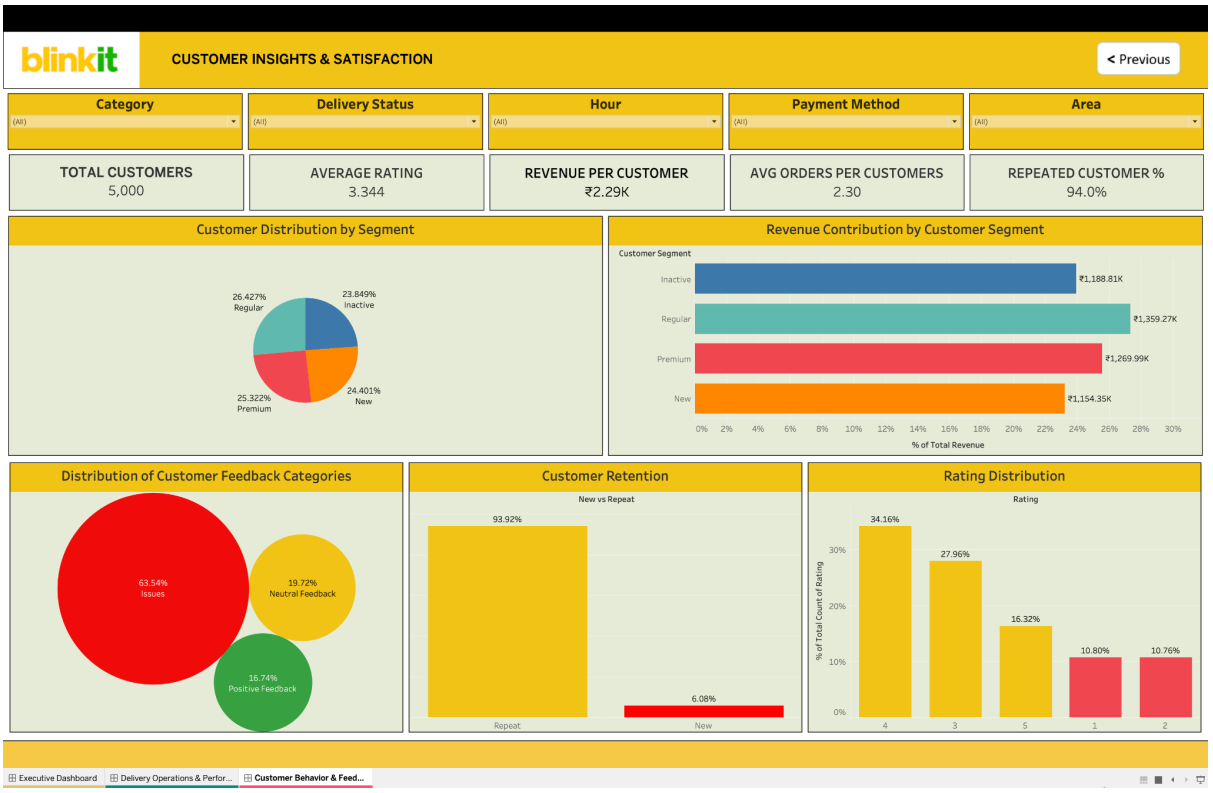
Shows revenue, categories, and payment trends

Delivery Dashboard:



Shows delay patterns, peak hours, slow stores, and partners

Customer Dashboard:



Shows customer segments, feedback, and ratings

These dashboards help both managers and analysts make decisions.

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10. Key Insights

1. Around 62% of orders are delayed
2. Delays are highest during peak hours
3. Few stores cause most delays
4. Some delivery partners perform poorly
5. Customer ratings are average (~3.3)
6. Around 63% feedback is negative
7. Revenue is strong despite issues
8. Digital payments are most used
9. Customer repeat rate is high
10. Problems are concentrated, not spread evenly

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11. Recommendations

1. Increase delivery capacity during peak hours
2. Improve performance of slow stores
3. Track delivery partner performance
4. Improve customer service
5. Optimize delivery routes

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12. Impact

If these changes are applied:

- Delivery delays will reduce
- Customer satisfaction will improve
- More customers will return
- Business performance will improve

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13. Limitations

- Data is not real-time
- Some assumptions were made
- External factors are missing

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14. Future Scope

- Use real-time data
- Build prediction models
- Improve delivery routes
- Study customer churn

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15. Conclusion

Blinkit is performing well in revenue but faces problems in delivery and customer satisfaction. By improving delivery operations, the company can improve customer experience and grow faster.

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16. Appendix

Data Dictionary

Below is a summary of key columns used in the analysis:

Column Name	Description
order_id	Unique identifier for each order
customer_id	Unique identifier for each customer
order_date	Date when the order was placed
order_total	Total value of the order
payment_method	Mode of payment (Card, Cash, UPI, Wallet)
delivery_partner_id	ID of the delivery partner
store_id	ID of the store fulfilling the order
hour	Hour of the day when order was placed
day / month / weekday	Time-based features for trend analysis
product_id	Unique identifier for product
category	Product category
quantity	Number of items ordered
unit_price	Price per item
distance_km	Distance for delivery
promised_time	Expected delivery time
actual_time	Actual delivery time
delivery_time_minutes	Total delivery duration
delay_minutes	Difference between actual and promised time
is_delayed	Indicates whether order was delayed
delay_bucket	Categorization of delay (0–5, 5–15, 15+ min)
rating	Customer rating (1–5)
feedback_text	Customer feedback comments
customer_segment	Segment (New, Regular, Premium, Inactive)

total_orders	Total orders by customer
avg_order_value	Average order value per customer

Additional EDA Outputs

Some additional analyses were performed but not included in the main dashboard:

- Distribution of order values across categories
- Relationship between delivery distance and delay
- Hour-wise order volume trends
- Category-wise delay comparison

These helped validate patterns but were excluded to keep the dashboard focused and simple.

Python Logic (ETL & Analysis)

Key steps from the data processing pipeline:

```
# Calculate delay
df['delay_minutes'] = df['actual_time'] - df['promised_time']

# Create delay flag
df['is_delayed'] = df['delay_minutes'] > 0

# Create delay buckets
def delay_category(x):
    if x <= 5:
        return '0-5 min'
    elif x <= 15:
        return '5-15 min'
    else:
        return '15+ min'

df['delay_bucket'] = df['delay_minutes'].apply(delay_category)

# Extract hour
df['hour'] = pd.to_datetime(df['order_date']).dt.hour
```

Cleaning Log Summary

The following steps were applied during data cleaning:

- Removed duplicate records
- Handled missing values using appropriate methods
- Converted date and time columns to proper format
- Standardized column names
- Removed inconsistent or invalid entries
- Created new features for analysis (delay, buckets, time features)

All cleaning steps are mentioned in the following:
ETL logs

Notes

- The dataset is simulated and may not fully represent real-world conditions
- Some assumptions were made while calculating delay
- External factors such as traffic and weather are not included

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16. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Swati Ghosh	✔	✔			✔	✔	✔
Aviral Mishra		✔	✔	✔			
Ishita Thakur						✔	✔
Tattva Rajput						✔	✔
Bhumi Rastog			✔				
Khuswant Rajpurohit			✔				

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artefacts.

Team Lead Signature: Swati Ghosh
Date: 29-04-2026